Probability theory and statistics

Lecture Notes

P&S team

Ukrainian Catholic University
BA & CS Programmes

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APPLIED SCIENCES FACULTY

Lecture 6. Continuous random variables

Outline

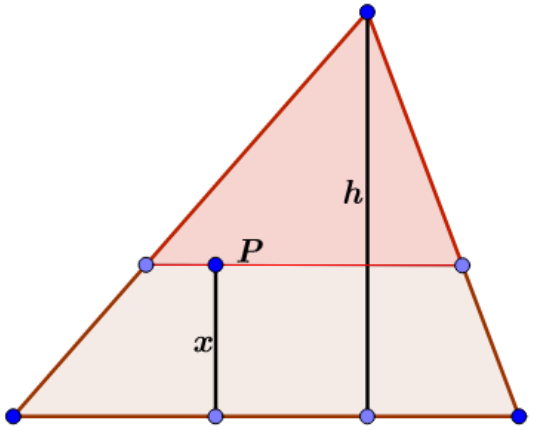
1. Probability density function
   - Continuous random variables
   - Properties of continuous r.v.
   - Expectation and variance
2. Continuous distributions and their properties
   - Uniform distribution
   - Normal, or Gaussian distribution
   - Exponential distribution
3. Joint distributions; transformation of random variables
   - Joint distributions
   - Independence
   - Covariance and correlation
   - Transformations
   - Conditional distributions

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Example

- A point P is randomly chosen inside a triangle Δ of height h and base a
- the distance from P to the base is measured and denoted X
- thus $\Omega = \{\text{interior points of } \Delta\}$;
  $\mathscr{F} = \{\text{“good” subsets of } \Omega\}$;
  P is a uniform measure (area)



- X is a **r.v.** on $(\Omega, \mathscr{F}, \mathrm{P})$
- **c.d.f.** (cumulative distribution function) of X:

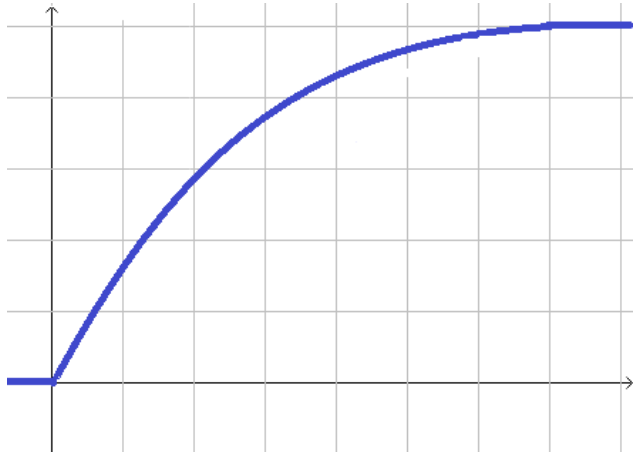
$$\mathrm{P}(X > x) = \frac{\mu(\Delta_1)}{\mu(\Delta)} = \left(\frac{h-x}{h}\right)^2$$

therefore,

$$F_X(x) = \mathrm{P}(X \le x) = \frac{x}{h}\left(2 - \frac{x}{h}\right), \qquad 0 \le x \le h$$

Probability density function

- The **c.d.f.** of X is continuous!
- $F_X(x) = 0$ for $x \leq 0$
  $F_X(x) = 1$ for $x \geq h$
- $\mathrm{P}(a < X \leq b) = F_X(b) - F_X(a)$
- $\mathrm{P}(\{X = a\}) = 0$

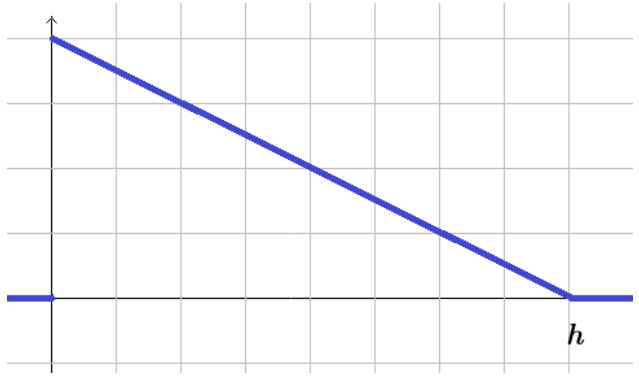


Definition (Probability density function)

If the **c.d.f.** F_X is differentiable (save finitely many pts), X is called continuous. Derivative $F'_X(x)$ is called the probability density function (**p.d.f.**) of the **r.v.** X and is denoted $f_X(x)$

Properties of **p.d.f.**:

- $f_X(x) \geq 0$;
- $\mathrm{P}(a < X \leq b) = \int_a^b f_X(t)\,dt$
- in particular, $\int_{\mathbb{R}} f_X(t)\,dt = 1$



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Properties of continuous **r.v.**

- $f_X \geq 0$ and $\int_{\mathbb{R}} f_X(x)\,dx = 1$ (the only requirements on **p.d.f.**!)

Example

Assume that $f_X(x) = 0$ for $x \leq a$ and $f_X(x) = c/x^2$ for $x \geq a$. What is c?
As $\int_{\mathbb{R}} f_X = 1$, we find that $c = a$

- For any $a \in \mathbb{R}$, $\mathrm{P}(\{X = a\}) = 0$:
  - $\mathrm{P}(\{a - \varepsilon < X \leq a + \varepsilon\}) = F_X(a + \varepsilon) - F_X(a - \varepsilon) \to 0 \quad \text{as} \quad \varepsilon \to 0$ since F_X is continuous at $x = a$
  - Therefore, the inclusion $\{X = a\} \subset \{a - \varepsilon < X \leq a + \varepsilon\}$ implies

$$\mathrm{P}(\{X = a\}) \leq \mathrm{P}(\{a - \varepsilon < X \leq a + \varepsilon\}) \to 0 \quad \text{as} \quad \varepsilon \to 0$$

- If $Y = aX + b$ with $a > 0$, then

$$F_Y(y) = \mathrm{P}(Y \leq y) = \mathrm{P}(X \leq (y - b)/a) = F_X\big((y - b)/a\big) \implies$$

$$f_Y(y) = F_Y'(y) = \tfrac{1}{a} f_X\big(\tfrac{y-b}{a}\big)$$

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Expectation

- For a discrete **r.v.**, $\mathrm{E}(X) = \sum_k x_k p_X(x_k)$
- For a continuous **r.v.**,

$$\boxed{\mathrm{E}(X) := \int_{\mathbb{R}} x\, f_X(x)\, dx}$$

provided $\int_{\mathbb{R}} |x|\, f_X(x)\, dx < \infty$.

Example

- In the above example with a triangle,

$$f_X(x) = F_X'(x) = \begin{cases} 0, & x \notin (0,h) \\ \frac{2}{h}(1 - \frac{x}{h}), & x \in (0,h) \end{cases}$$

- therefore,

$$\mathrm{E}(X) = \int_{\mathbb{R}} x\, f_X(x)\, dx = 2\int_0^h \tfrac{x}{h}(1 - \tfrac{x}{h})\, dx = \tfrac{x^2}{h}\Big|_0^h - \tfrac{2x^3}{3h^2}\Big|_0^h = \tfrac{h}{3}$$

Example

Example

If $f_X(x) = a/x^2$ for $x \geq a > 0$ and $f_X(x) = 0$ otherwise, then

$$F_X(x) = \int_a^x as^{-2}\,ds = 1 - a/x \quad \text{for} \quad x > a;$$

$$\mathrm{E}(X) = \int_{\mathbb{R}} x\, f_X(x)\,dx = \int_a^\infty \frac{a\,dx}{x} = \infty$$

thus X does not have finite expectation

Example

Example (Standard normal distribution)

A **r.v.** with **p.d.f.** $f(x) = e^{-x^2/2}/\sqrt{2\pi}$ is said to have a standard normal distribution. Observe that indeed (use e.g. the Gamma-function)

$$\int_{\mathbb{R}} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = 1$$

Since the density is an even function,

$$\mathrm{E}(X) = \int_{\mathbb{R}} x\, f(x)\, dx = 0$$

Properties of expectation

- Expectation of transformed **r.v.**: For a function $g : \mathbb{R} \to \mathbb{R}$,
$$\mathrm{E}(g(X)) := \int_{\mathbb{R}} g(x)\, f_X(x)\, dx$$
- Linearity: $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$:
$$\mathrm{E}(aX+b) = \int_{\mathbb{R}} (ax+b) f_X(x)\, dx = \int_{\mathbb{R}} ax\, f_X(x) + \int_{\mathbb{R}} b f_X(x) = a\mathrm{E}(X)+b$$
- In general, $\mathrm{E}((g(X)) \neq g(\mathrm{E}(X))$;
for convex g, Jensen's inequality gives the "$\geq$" sign:
  - $\mathrm{E}(|X|) \geq |\mathrm{E}(X)|$
  - $\mathrm{E}(X^2) \geq \mathrm{E}^2(X)$ etc

Example (Triangle)

Set $Y := \frac{1}{h}X$; then $f_Y(y) = h f_X(hy) = 2(1-y)$ for $y \in (0,1)$ and zero otherwise;
$$\mathrm{E}(Y) = \int_0^1 2y(1-y)\, dy = 1 - \tfrac{2}{3} = \tfrac{1}{3} = \tfrac{1}{h}\mathrm{E}(X)$$

Moment generating functions

Definition

For a continuous (or discrete) **r.v.** X, its moment generating function (**m.g.f.**) $M_X(t)$ is

$$M_X(t) = \mathrm{E}(e^{tX}) = \int_{\mathbb{R}} e^{tx} f_X(x)\, dx$$

Example

For X with the standard normal distribution,

$$M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{tx} e^{-x^2/2}\, dx = e^{t^2/2}$$

Moment generating functions

Why m.g.f.?

$$M'_X(t) = \mathrm{E}(X\, e^{tX}) \text{ so that } M'_X(0) = \mathrm{E}(X);$$

$$M''_X(t) = \mathrm{E}(X^2\, e^{tX}) \text{ so that } M''_X(0) = \mathrm{E}(X^2)$$

$$M^{(k)}_X(t) = \mathrm{E}(X^k\, e^{tX}) \text{ so that } M^{(k)}_X(0) = \mathrm{E}(X^k)$$

Remark

M.g.f. is a contiunous analogue of the p.g.f., with similar properties

Properties of **m.g.f**:

- $M^{(k)}_X(0) = \mathrm{E}(X^k)$
- If X and Y are independent, then $M_{X+Y}(s) = M_X(s)M_Y(s)$
- Moment generating function uniquely determines the distribution.

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Uniform distribution

Definition (Uniform distribution)

A continuous random variable X has a uniform distribution $\mathscr{U}(a, b)$ in an interval (a, b) (written $X \sim \mathscr{U}(a, b)$) if its density is

$$f_X(t) = \frac{1}{b-a}\chi_{(a,b)}(t) = \begin{cases} \frac{1}{b-a}, & t \in (a, b) \\ 0 & \text{otherwise.} \end{cases}$$

- $X \sim \mathscr{U}(a, b)$ takes values in (a, b) only;
- $\mathrm{P}(X \in \Delta)$ only depends on the size of Δ;
- the expectation $\mathrm{E}X$ is

$$\mathrm{E}X = \frac{1}{b-a}\int_a^b t\,dt = \frac{1}{2(b-a)}t^2\Big|_{t=a}^{b} = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$$

and is the middle point of the interval (a, b)

- the second moment is

$$\mathrm{E}(X^2) = \frac{1}{b-a}\int_a^b t^2\,dt = \frac{1}{3(b-a)}t^3\Big|_{t=a}^{b}$$
$$= \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3},$$

- the variance is

$$\mathrm{Var}X = \mathrm{E}(X^2) - \mathrm{E}^2(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4}$$
$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$
$$= \frac{(b-a)^2}{12}$$

- It is natural that the variance is proportional to $(b-a)^2$!

Example

To calculate the above values in another way, write

$$X = \frac{a+b}{2} + \frac{b-a}{2} U;$$

then U has a uniform distribution over $(-1, 1)$.
By symmetry, the expectation of U is zero ($\mathrm{E}\,U = 0$), whence

$$\operatorname{Var} U = \mathrm{E}(U^2) = \frac{1}{2} \int_{-1}^{1} t^2 \, dt = \frac{1}{3}.$$

By linearity of expectation, we find that

$$\mathrm{E}\, X = \frac{a+b}{2} + \frac{b-a}{2} \mathrm{E}\, U = \frac{a+b}{2};$$

since $\operatorname{Var}(cU) = c^2 \operatorname{Var}(U)$ and variance of a constant is zero,

$$\operatorname{Var} X = \frac{(b-a)^2}{4} \operatorname{Var} U = \frac{(b-a)^2}{12}.$$

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Normal, or Gaussian distribution

Definition (Normal, or Gaussian distribution)

A standard normal, or Gaussian, distribution $\mathcal{N}(0,1)$ is given by the density function

$$\phi(t) := \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2}$$

Its c.d.f. is usually denoted Φ:

$$\Phi(t) := \int_{-\infty}^{t} \phi(s)\, ds = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} \mathrm{e}^{-s^2/2}\, ds$$

If X has distribution $\mathcal{N}(0,1)$ (written $X \sim \mathcal{N}(0,1)$), then its expectation is zero and the variance is 1:

$$\mathrm{E}\, X = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} t \mathrm{e}^{-t^2/2}\, dt = 0,$$

$$\mathrm{Var}\, X = \mathrm{E} X^2 = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} t^2 \mathrm{e}^{-t^2/2}\, dt = 1.$$

To prove that $\mathrm{E}(X^2) = 1$:

- use parity to get

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} t^2 \mathrm{e}^{-t^2/2}\, dt = \sqrt{\frac{2}{\pi}} \int_0^\infty t^2 \mathrm{e}^{-t^2/2}\, dt$$

- make substitution $u = t^2/2$, or $t = \sqrt{2u}$
- conclude that $dt = du/\sqrt{2u}$ and

$$\sqrt{\frac{2}{\pi}} \int_0^\infty t^2 \mathrm{e}^{-t^2/2}\, dt = \frac{2}{\sqrt{\pi}} \int_0^\infty \sqrt{u} \mathrm{e}^{-u}\, du = \frac{2}{\sqrt{\pi}} \cdot \Gamma\left(\tfrac{3}{2}\right) = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1$$

in view of the relations $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

- here $\Gamma(\cdot)$ is the Euler Gamma-function

$$\Gamma(s) := \int_0^\infty t^{s-1} e^{-t}\, dt;$$

observe that $\Gamma(n+1) = n!$

Alternatively, use the m.g.f. M_X:

$$M_X(t) := \mathrm{E}(e^{tX}) = \int_{\mathbb{R}} e^{tx}\phi(x)\,dx = e^{t^2/2}$$

thus

$$m_1(X) = \mathrm{E}(X) = M'_X(0) = \left(te^{t^2/2}\right)\Big|_{t=0} = 0$$

$$m_2(X) = \mathrm{E}(X^2) = M''_X(0) = \left([t^2+1]e^{t^2/2}\right)\Big|_{t=0} = 1$$

$$m_3(X) = \mathrm{E}(X^3) = M'''_X(0) = \left([t^3+3t]e^{t^2/2}\right)\Big|_{t=0} = 0$$

$$m_4(X) = \mathrm{E}(X^4) = M^{(iv)}_X(0) = \left([t^4+6t^2+3]e^{t^2/2}\right)\Big|_{t=0} = 3$$

and so on

Definition

If $Z \sim \mathcal{N}(0,1)$, then the random variable $\mu + \sigma Z$ is said to have a normal (or Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$.

- If $Y = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$, then E $Y = \mu$ and Var $Y = \sigma^2$
- this explains the rôle of the parameters μ and σ^2
- the density of Y:

$$F_Y(t) = \mathrm{P}(Y \le t) = \mathrm{P}(Z \le (t-\mu)/\sigma) = F_Z\left(\tfrac{t-\mu}{\sigma}\right),$$

so that

$$\boxed{F_Y(t) = \Phi\left(\tfrac{t-\mu}{\sigma}\right)}$$

and

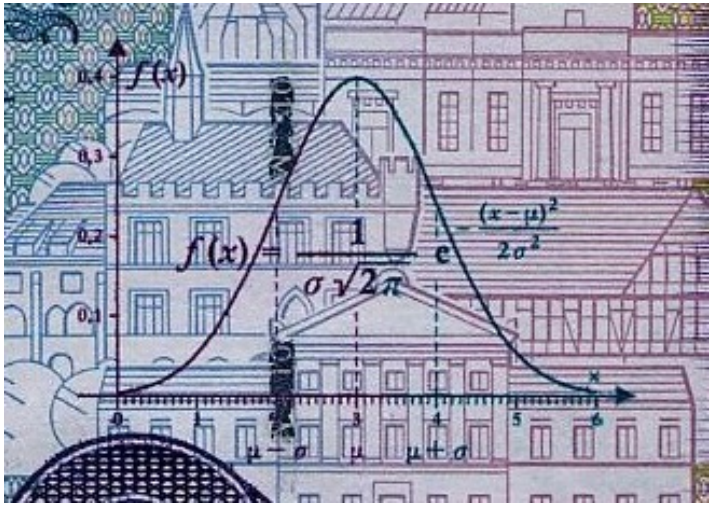
$$f_Y(t) = F_Y'(t) = \frac{1}{\sigma}\phi\left(\tfrac{t-\mu}{\sigma}\right),$$

i.e.,

$$\boxed{f_Y(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(t-\mu)^2}{2\sigma^2}\right\}}.$$

Normal distribution





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Definition (Exponential distribution)

A nonnegative random variable X has exponential distribution with parameter $\lambda > 0$, written $X \sim \mathcal{E}(\lambda)$, if the density of its distribution is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Example (Expectation of exponential distribution)

For $X \sim \mathcal{E}(\lambda)$, we make change of variables $\lambda x = y$ and integrate by parts to get

$$\begin{aligned} \mathrm{E}\, X &= \int_0^\infty x\lambda e^{-\lambda x}\, dx = \frac{1}{\lambda}\int_0^\infty y e^{-y}\, dy = -\frac{1}{\lambda}\int_0^\infty y(e^{-y})'\, dy \\ &= y e^{-y}\Big|_{y=0}^{\infty} + \frac{1}{\lambda}\int_0^\infty e^{-y}\, dy = \frac{1}{\lambda}; \end{aligned}$$

alternatively,

$$= \frac{1}{\lambda}\int_0^\infty y e^{-y}\, dy = \frac{1}{\lambda}\Gamma(2) = \frac{1!}{\lambda} = \frac{1}{\lambda}.$$

Example (Exponential distribution (cont.))

Recalling the definition of the Gamma function, $\Gamma(s) = \int_0^\infty t^{s-1}\mathrm{e}^{-t}\,dx$, *and the relation* $\Gamma(k) = (k-1)!$ *for natural* k, *we also find that*

$$\begin{aligned}\mathrm{E}\,X^k &= \int_0^\infty x^k \lambda \mathrm{e}^{-\lambda x}\,dx = \frac{1}{\lambda^k}\int_0^\infty y^k \mathrm{e}^{-y}\,dy \\ &= \frac{\Gamma(k+1)}{\lambda^k} = \frac{k!}{\lambda^k};\end{aligned}$$

in particular,

$$\mathrm{Var}X = \mathrm{E}(X^2) - (\mathrm{E}\,X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Alternatively, find the m.g.f. M_X*:*

$$M_X(t) = \int_0^\infty e^{tx} f_X(x)\,dx = \int_0^\infty \lambda e^{-x(\lambda - t)}\,dt = \frac{\lambda}{\lambda - t}$$

Lack of memory property:

$$\mathrm{P}(X > s + t \mid X > s) = \mathrm{P}(X > t)$$

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Joint distributions

Definition (Joint c.d.f.)

Assume X and Y are **r.v.** defined on a probability space $(\Omega, \mathscr{F}, \mathrm{P})$. Their joint cumulative distribution function $F_{X,Y}$ is

$$F_{X,Y}(x, y) := \mathrm{P}(X \leq x, Y \leq y)$$

The marginal c.d.f. of X and Y are

$$F_X(x) = \lim_{y \to \infty} F_{X,Y}(x, y)$$

$$F_Y(y) = \lim_{x \to \infty} F_{X,Y}(x, y)$$

Example (Uniform distribution on the unit square)

$$F_{X,Y} = \begin{cases} 1 & x > 1, y > 1 \\ x, & x \in (0,1), y > 1 \\ y, & x > 1, y \in (0,1) \\ xy, & x, y \in (0,1) \\ 0, & \text{otherwise} \end{cases}$$

Joint probability density function

Definition

We say that **r.v.** X and Y have continuous joint distribution if

$$F_{X,Y}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,du\,dv$$

for a nonnegative joint density function $f_{X,Y}$. In that case,

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy$$

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dx$$

give marginal densities of X and Y

Example

The uniform distribution on the unit square S has a joint **p.d.f.**

$$f_{X,Y}(x,y) = \begin{cases} 1 & \text{on } S \\ 0 & \text{otherwise} \end{cases}$$

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Independence

Definition

R.v.'s X and Y are called independent if for all x and y the events $\{X \leq x\}$ and $\{Y \leq y\}$ are independent.

- Thus for independent X and Y

$$\mathrm{P}(X \leq x, Y \leq y) = \mathrm{P}(X \leq x) \cdot \mathrm{P}(Y \leq y);$$

- or, in terms of cumulative distribution functions (**c.d.f.**)

$$F_{X,Y}(x, y) = F_X(x) F_Y(y)$$

- or, in terms of probability density functions (**p.d.f.**),

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y).$$

Remark

The notion of independence is extended in a natural way to any number of random variables.

Some characteristics of joint distributions

- For any **r.v.**'s X and Y, $\mathrm{E}(X+Y) = \mathrm{E}\,X + \mathrm{E}\,Y$
- Variance of a **r.v.** X:
$$\mathrm{Var}\,X := \mathrm{E}\big((X-\mathrm{E}X)^2\big) = \mathrm{E}(X^2) - (\mathrm{E}X)^2$$
- Additivity of variances
$$\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y \tag{1}$$
cannot hold for all **r.v.**'s (take $Y = -X$ for a nonconstant X), but it holds for independent **r.v.**'s.
- X and Y independent $\Longrightarrow$
$$\mathrm{E}(XY) = \mathrm{E}\,X \cdot \mathrm{E}\,Y$$
and thus
$$\mathrm{E}\big((X+Y)^2\big) = \mathrm{E}(X^2 + 2XY + Y^2) = \mathrm{E}(X^2) + 2\mathrm{E}X \cdot \mathrm{E}\,Y + \mathrm{E}(Y^2),$$
- therefore, for independent X and Y
$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathrm{E}\big((X+Y)^2\big) - \big(\mathrm{E}(X+Y)\big)^2 \\ &= \mathrm{E}(X^2) - (\mathrm{E}X)^2 + \mathrm{E}(Y^2) - (\mathrm{E}\,Y)^2 = \mathrm{Var}\,X + \mathrm{Var}\,Y.\end{aligned}$$

Independent **r.v.**'s

X and Y independent $\implies$ $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ and

$$\mathrm{E}(XY) = \int_{\mathbb{R}}\int_{\mathbb{R}} xy\, f_{X,Y}(x,y)\,dx\,dy = \int_{\mathbb{R}}\int_{\mathbb{R}} xy\, f_X(x)f_Y(y)\,dx\,dy$$
$$= \int_{\mathbb{R}} xf_X(x)\,dx \cdot \int_{\mathbb{R}} yf_Y(y)\,dy = \mathrm{E}\,X \cdot \mathrm{E}\,Y$$

as claimed. More generally,

$$\mathrm{E}\big(g(X)h(Y)\big) = \mathrm{E}\big(g(X)\big)\mathrm{E}\big(h(Y)\big)$$

for any continuous functions g and h.

Remark

- *The equality* $\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y$ *does not imply that* X *and* Y *are independent*
- *Also, the above equality extends to any number of pairwise independent summands* $X_1, X_2, \ldots, X_n$*, i.e.,*

$$\mathrm{Var}(X_1 + X_2 + \cdots + X_n) = \mathrm{Var}\,X_1 + \mathrm{Var}\,X_2 + \cdots + \mathrm{Var}\,X_n.$$

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Covariance and correlation

Definition (Covariance)

For two random variables X and Y, their covariance $\mathrm{Cov}(X, Y)$ is

$$\mathrm{Cov}(X, Y) := \mathrm{E}\Big((X - \mathrm{E}X) \cdot (Y - \mathrm{E}Y)\Big) = \mathrm{E}(XY) - \mathrm{E}X \cdot \mathrm{E}Y.$$

If $\mathrm{Cov}(X, Y) = 0$, the r.v. X and Y are called uncorrelated.

- Independent X and Y are uncorrelated since $\mathrm{E}(XY) = \mathrm{E}\,X \cdot \mathrm{E}\,Y$
- The converse is not true: e.g. $X = Z$ and $Y = Z^2$ for standard Gaussian Z are uncorrelated but dependent
- For **r.v.**'s X and Y,

$$\mathrm{Var}(X + Y) = \mathrm{Var}X + 2\mathrm{Cov}(X, Y) + \mathrm{Var}\,Y$$

- In particular, for independent X and Y

$$\mathrm{Var}(X + Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y$$

- $\mathrm{Cov}(X, Y) > 0$ means that larger values of X correspond to larger values of Y

Covariance and correlation

Definition (Correlation)

For two random variables X and Y, their correlation $\mathrm{Cor}(X, Y)$ is

$$\mathrm{Cor}(X, Y) := \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

- For any two **r.v.**'s X and Y,

$$|\mathrm{Cor}(X, Y)| \leq 1$$

  (follows from Cauchy–Bunyakowski–Schwarz inequality)
- For independent X and Y, $\mathrm{Cor}(X, Y) = 0$
- $\mathrm{Cor}(X, Y) = 0 \implies X$ and Y are uncorrelated
  (but need not be independent)
- If $\mathrm{Cor}(X, Y) = 1$, then $Y = aX + b$ with $a > 0$
- If $\mathrm{Cor}(X, Y) = -1$, then $Y = aX + b$ with $a < 0$

Outline

1. Probability density function
   - Continuous random variables
   - Properties of continuous r.v.
   - Expectation and variance
2. Continuous distributions and their properties
   - Uniform distribution
   - Normal, or Gaussian distribution
   - Exponential distribution
3. Joint distributions; transformation of random variables
   - Joint distributions
   - Independence
   - Covariance and correlation
   - Transformations
   - Conditional distributions

Distribution of transformed variables

Question:

If $Y = g(X)$ for some function g, what is the distribution of Y?

$Y = aX + b$ for $a > 0$ $\Longrightarrow$

$$F_Y(y) = \mathrm{P}(Y \leq y) = \mathrm{P}(aX+b \leq y) = \mathrm{P}(X \leq (y-b)/a) = F_X((y-b)/a)$$

and the **p.d.f.** of Y is

$$f_Y(y) = F'_Y(y) = \tfrac{1}{a} F'_X((y-b)/a) = \tfrac{1}{a} f_X((y-b)/a)$$

More generally, if $Y = g(X)$ for an increasing function g, then

$$F_Y(y) = \mathrm{P}(Y \leq y) = \mathrm{P}(g(X) \leq y) = \mathrm{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

and the **p.d.f.** of Y is

$$f_Y(y) = F'_Y(y) = (g^{-1})'(y) f_X(g^{-1}(y))$$

Example

If $X \sim \mathscr{U}(0,1)$ and $Y = -\ln X$, then $Y \sim \mathscr{E}(1)$

Distribution of transformed variables

Assume that

- $\mathbf{X} = (X_1, \ldots, X_n)$ is a continuous random vector;
- $S \subset \mathbb{R}^n$ is open and $\mathrm{P}(\mathbf{X} \in S) = 1$;
- $g : S \to \mathbb{R}^n$ is one-to-one, continuously differentiable and non-degenerate on S.

Then the random vector $\mathbf{Y} := g(\mathbf{X})$ is continuously distributed with

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}\big(\mathbf{g}^{-1}(\mathbf{y})\big) \cdot |J_{\mathbf{g}^{-1}}(\mathbf{y})| = f_{\mathbf{X}}\big(\mathbf{g}^{-1}(\mathbf{y})\big) / |J_{\mathbf{g}}\big(\mathbf{g}^{-1}(\mathbf{y})\big)|.$$

($J_{\mathbf{h}}$ is the Jacobian of mapping $\mathbf{h}$, i.e., det. of the matrix $\big(\partial h_j / \partial y_k\big)_{j,k=1}^{n}$).

Proof is based on the change of variables formula $\mathbf{x} = g^{-1}(\mathbf{y})$:

$$F_{\mathbf{Y}}(\mathbf{t}) = \mathrm{P}\{\mathbf{g}(\mathbf{X}) \le \mathbf{t}\} = \int_{\mathbf{g}(\mathbf{x}) \le \mathbf{t}} f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x}$$

$$= \int_{\mathbf{y} \le \mathbf{t}} f_{\mathbf{X}}\big(\mathbf{g}^{-1}(\mathbf{y})\big) |J_{\mathbf{g}^{-1}}(\mathbf{y})|\, d\mathbf{y} = \int_{\mathbf{y} \le \mathbf{t}} f_{\mathbf{Y}}(\mathbf{y})\, d\mathbf{y}$$

Examples

Scalar case: $Y = g(X)$

$$f_Y(y) = f_X\big(g^{-1}(y)\big) \cdot |(g^{-1})'(y)| = f_X\big(g^{-1}(y)\big) / |g'\big(g^{-1}(y)\big)|$$

If $X \sim \mathscr{E}(1)$ and $Y = e^{-X}$, then

$$f_Y(y) = \tfrac{1}{y} f_X(-\log y) = \tfrac{1}{y} e^{\log y} = 1, \qquad y = e^{-x} \in (0,1)$$

Affine transformation $\mathbf{g}(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$

$J_{\mathbf{g}} = \det A$ and $J_{\mathbf{g}^{-1}} = \det(A^{-1}) = 1/\det A$, so that

$$f_{\mathbf{Y}}(\mathbf{y}) = |\det A|^{-1} f_{\mathbf{X}}\big(A^{-1}(\mathbf{y} - \mathbf{b})\big)$$

For example, $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$ is the mapping

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \qquad A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

thus

$$f_{\mathbf{Y}}(\mathbf{y}) = \tfrac{1}{2} f_{\mathbf{X}}\left(\tfrac{y_1 + y_2}{2}, \tfrac{y_1 - y_2}{2} \right)$$

Distribution of $X_1 + X_2$ and convolution

Distribution of $X_1 + X_2$

Consider $Y_1 = X_1 + X_2$ and $Y_2 = X_2$; then $g^{-1}(y_1, y_2) = (y_1 - y_2, y_2)$ and

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(y_1 - y_2, y_2)$$

and the marginal **p.d.f.** of Y_1 (i.e., **p.d.f.** of $X_1 + X_2$) is

$$f_{X_1+X_2}(y_1) = f_{Y_1}(y_1) = \int_{\mathbb{R}} f_{\mathbf{X}}(y_1 - t, t)\, dt = \int_{\mathbb{R}} f_{\mathbf{X}}(t, y_1 - t)\, dt$$

If X_1 and X_2 are independent, then we get the convolution of f_{X_1} and f_{X_2}:

$$\boxed{f_{X_1+X_2}(y) = \int_{\mathbb{R}} f_{X_1}(y - t) f_{X_2}(t)\, dt = \int_{\mathbb{R}} f_{X_1}(t) f_{X_2}(y - t)\, dt}$$

Bertrand paradox:

Problem

A chord on a given circle of radius 1 *is chosen at random. What is the probability that its length is greater than the side length of the equilateral triangle inscribed in the circle?*

Solution

The answer is $\frac{1}{4}$ *because the chord centre must lie in the concentric circle of radius* $\frac{1}{2}$

Solution

The answer is $\frac{1}{3}$ *because if one point is fixed, the other should lie in the middle third of the circumference*

Solution

The answer is $\frac{1}{2}$ *because if the chord direction is fixed, its center must lie on the middle half of the perpendicular diameter*

Bertrand paradox

- Choose points A and B independently and uniformly on the unit circumference;
- A and B can be described by the angles $\alpha, \beta \sim \mathscr{U}[0, 2\pi)$;
- chord AB has length at least $\sqrt{3}$ iff $2\pi/3 \leq |\alpha - \beta| \leq 4\pi/3$
- the probability of this is $1/3$
- Alternatively: choose the middle point C of AB
- C has coordinates $X = \frac{1}{2}(\cos\alpha + \cos\beta)$ and $Y = \frac{1}{2}(\sin\alpha + \sin\beta)$
- C is not uniformly distributed in the unit disc!
- Alternatively: find the direction $\gamma := \frac{1}{2}(\alpha + \beta) \in [0, 2\pi)$ and the distance r to C from the centre
- r is equal to $\cos\frac{1}{2}(\alpha - \beta)$ and is not uniformly distributed on $(0, 1)$
- $f_R(r) = \frac{2}{\pi}\arccos'(r) = 2/\pi\sin(\arccos r) = 2/\pi\sqrt{1 - r^2}$; then

$$\mathrm{P}(R \leq \tfrac{1}{2}) = \int_0^{1/2} \frac{2dr}{\pi\sqrt{1-r^2}} = \frac{2}{\pi}\arcsin r\Big|_{r=0}^{1/2} = \frac{2}{\pi}\frac{\pi}{6} = \frac{1}{3}$$

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Conditional distribution

- Let X and Y have joint pdf $f_{X,Y}$;
- for small intervals Δ_x and Δ_y around x and y, we have

$$\mathrm{P}(X \in \Delta_x, Y \in \Delta_y) \approx f_{X,Y}(x,y)|\Delta_x||\Delta_y|$$

- thus

$$\mathrm{P}(X \in \Delta_x \mid Y \in \Delta_y) \approx \frac{f_{X,Y}(x,y)|\Delta_x|}{f_Y(y)}$$

- in the limit as $\Delta_y \to \{y\}$, we find that

$$\mathrm{P}(X \in \Delta_x \mid Y = y) \approx \frac{f_{X,Y}(x,y)}{f_Y(y)}|\Delta_x|$$

Definition

Given a joint **p.d.f.** $f_{X,Y}$, we say that the

conditional distribution of X given that $Y = y$

is continuous with the **p.d.f.**

$$f_{X|Y}(x \mid y) := \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Conditional distribution

Let X_1 and X_2 be independent with distribution $\mathscr{E}(\lambda)$

- Take $X = X_1$ and $Y = X_1 + X_2$;
- then $f_{X,Y}(x, y) = \lambda^2 e^{-\lambda y}$ on the set $0 < x < y$;
- $f_Y(y) = \int_0^y f_{X,Y}(x, y)\, dx = \lambda^2 y e^{-\lambda y}$;
- i.e., Y has the Gamma distribution $\Gamma(2, \lambda)$;
- now $f_{X|Y}(x \mid y) = y^{-1}$ for $x \in (0, y)$, so that $X \mid Y = y$ is a uniform distribution $\mathscr{U}(0, y)$
- clearly, $X \sim \mathscr{E}(\lambda)$, so that $f_{Y|X}(y \mid x) = \lambda e^{-\lambda(y-x)}$
- the distribution of $Y \mid X = x$ is shifted exponential i.e., $Y - x \mid X = x \sim \mathscr{E}(\lambda)$
- In fact, $Z := Y - X$ and X are independent, so that $(Y - x \mid X = x) \sim (Z \mid X = x) \sim Z$